

Understanding Agile

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Welcome

- **Ground rules**
- **Training Overview**
- **Why Agile**
- **Agile History**
- **Agile Manifesto**
- **Scrum**
- **Scrum Values**
- **Scrum events**
- **Product back**

What is Agile

- Developed for Software projects, but it is a methodology that can be used on all Projects types
- Agile is an umbrella term that is used to refer to different types of iterative development
- Scrum is the most common method of agile, there are others such as extreme programming (XP), lean development, and Kanban.

Waterfall v/s Agile

Waterfall

Requirements

Design

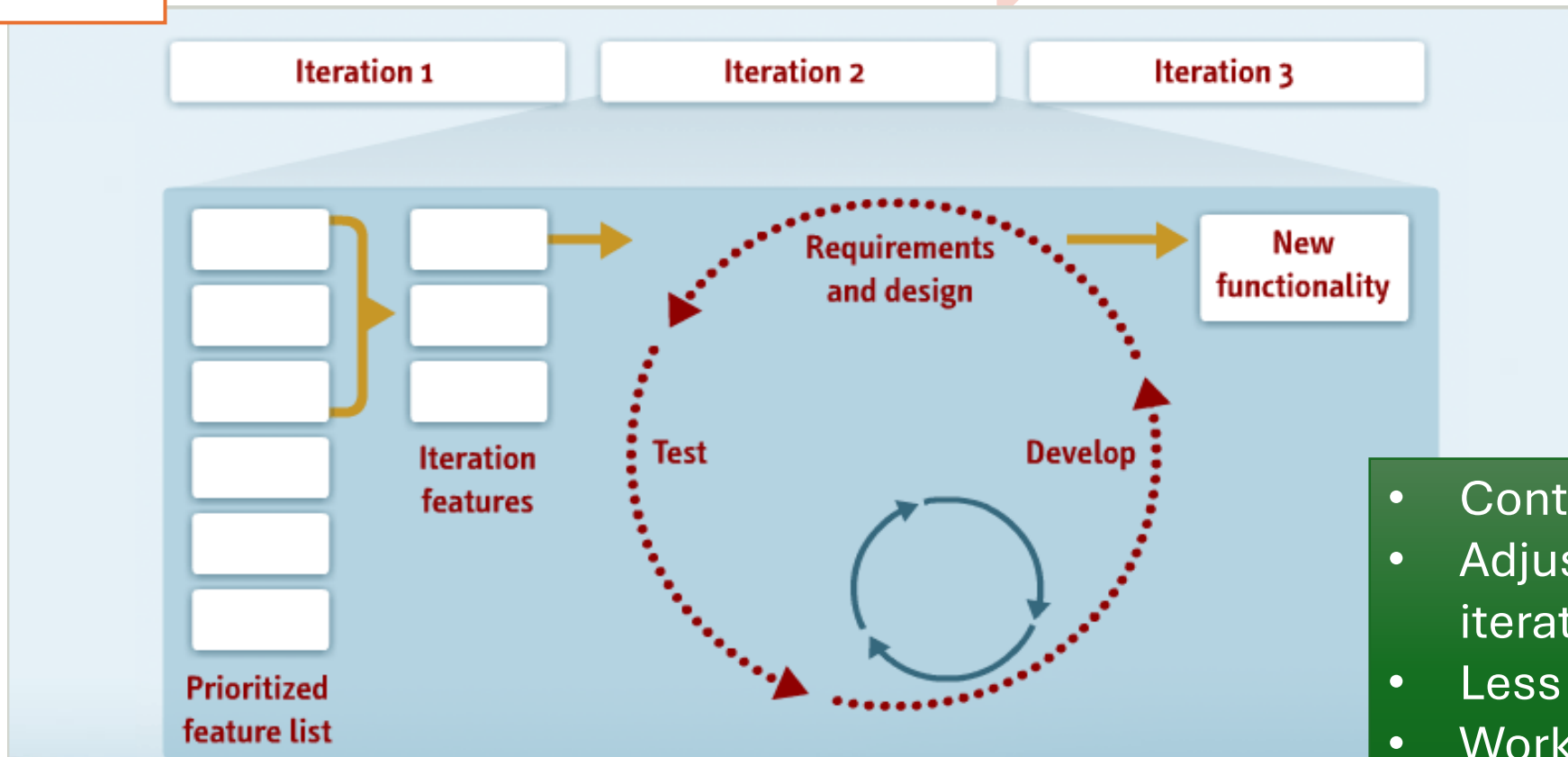
Development

Testing

Deployment

- Feedback is late
- Errors are Identified late
- Changes are costly

Agile



- Continuous Feedback
- Adjustments in each iteration
- Less Cost of changes
- Working software is the focus

Agile vs. Traditional Project Management

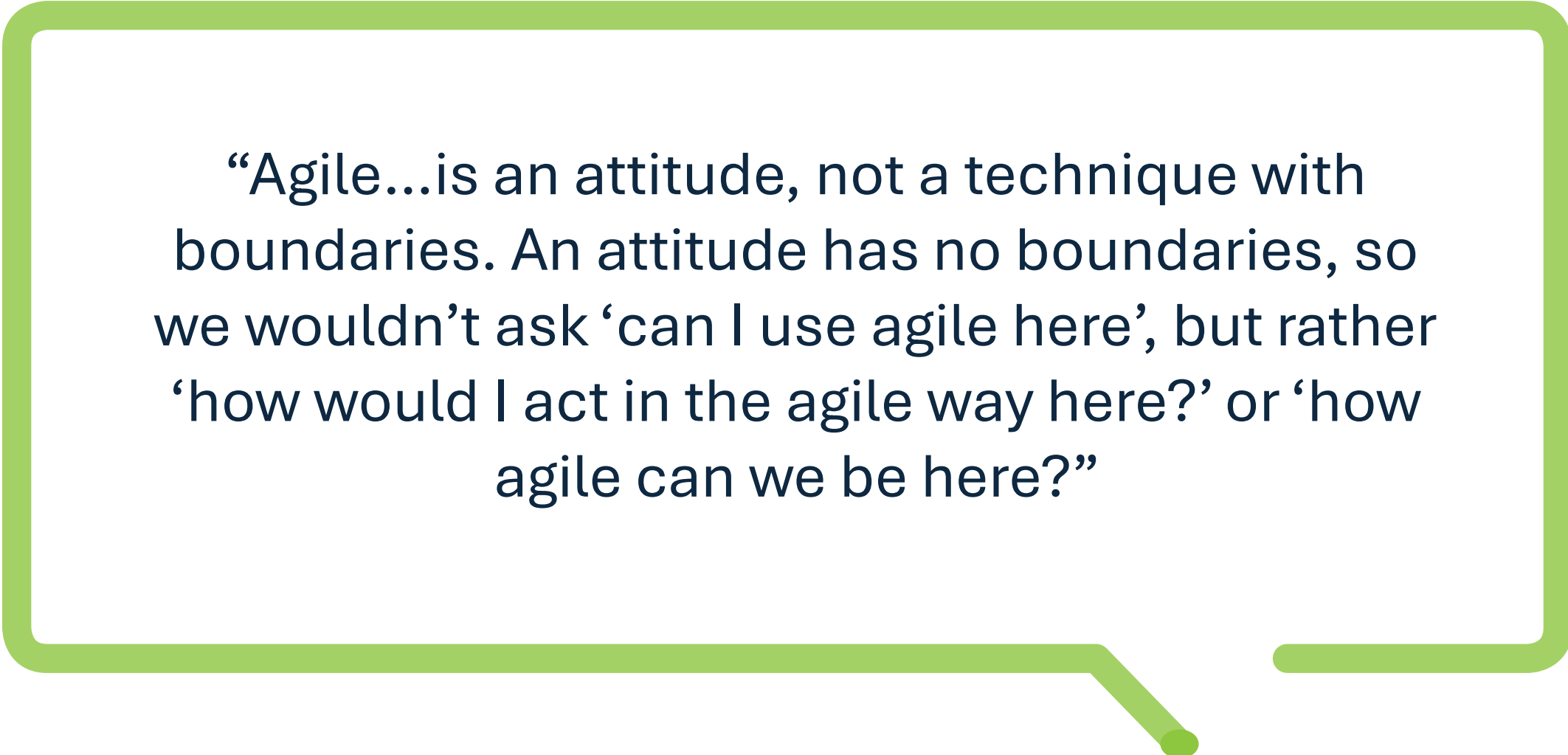
- Agile builds in increments vs. as a whole
- Agile does planning throughout vs. done all at once
- Agile delivers products over time vs. all at once
- Customers sees value faster vs. at the end
- Agile wants changes vs. discouraging changes

Agile Benefits

- Customer involved throughout the life cycle
- Greater Customer Interaction with all stakeholders
- Constant Feedback is required to stay current and successful
- Greater Value up front
- Change is welcomed by all stakeholders

Agile mindset

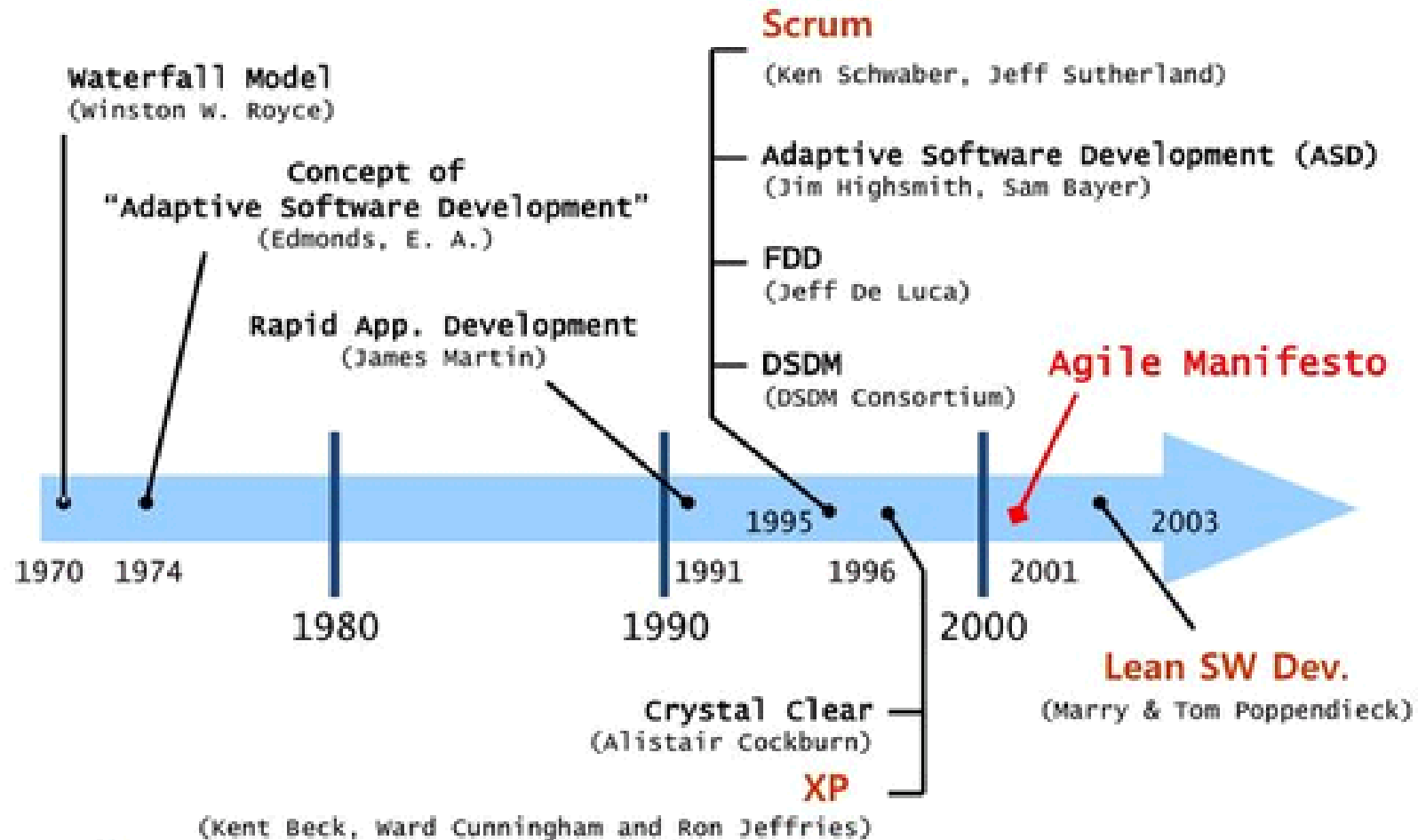
- Welcoming change
- Working in small value increments
- Using build and feedback loops
- Learning through discovery
- Value -driven development
- Failing fast with learning
- Continuous delivery
- Continuous improvement



“Agile...is an attitude, not a technique with boundaries. An attitude has no boundaries, so we wouldn't ask ‘can I use agile here’, but rather ‘how would I act in the agile way here?’ or ‘how agile can we be here?’”

— Alistair Cockburn

History of Agile



The Agile Manifesto - a statement of values

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals & interactions	over	Processes & tools
Working software	over	Comprehensive documentation
Customer collaboration	over	Contract negotiation
Responding to change	over	Following a plan

That is, while there is value in the items on the **right**, we value the items on the **left** more.

www.agilemanifesto.org

Twelve Principles of Agile

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
4. Business people and developers must work together daily throughout the project.
5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
6. Working software is the primary measure of progress.
7. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity--the art of maximizing the amount of work not done--is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

4 Pillars of Agile



When agile methodology was introduced?

- A. 2001
- B. 2009
- C. 2000
- D. 2004

According to Agile Manifesto

- A. Individuals and interactions over people and technique
- B. Individuals and interactions over processes and tools
- C. Individuals and interactions over projects and tools
- D. Individuals and interactions over products and tools

Which of the following is the working culture of an Agile Team?

- A. Connective
- B. Contemplative
- C. Collaborative
- D. Individual Contributors

There is no difference between incremental and iterative development

- A. True
- B. False

Which of the following is not a principle of the Agile Manifesto?

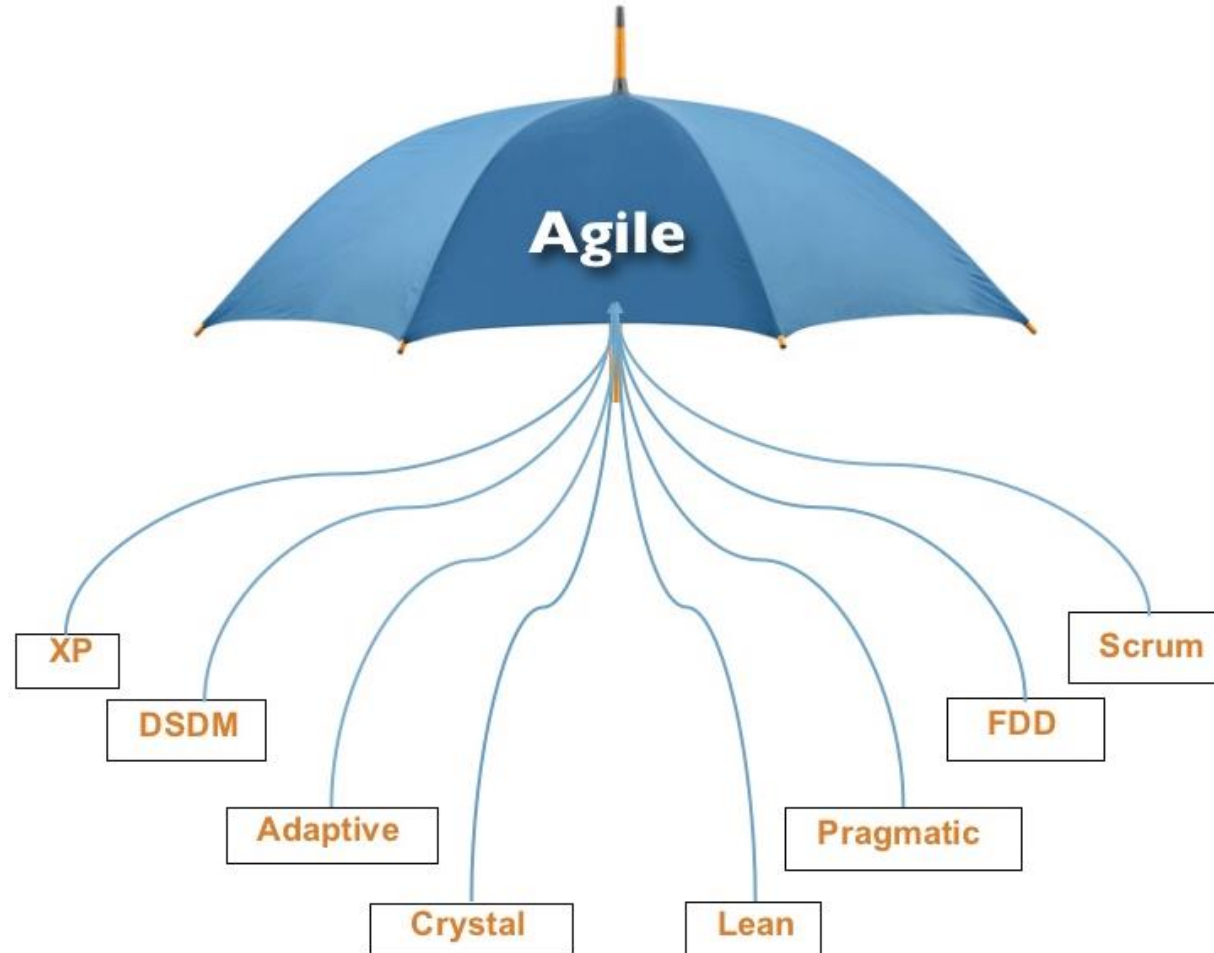
- A. Commitment to the plan despite changing circumstances
- B. Projects are built around motivated individuals, who should be trusted
- C. Working solution is delivered frequently (weeks rather than months)
- D. Face-to-face conversation is the best form of communication (co-location)

Agile Manifesto says to value responding to change over following a plan. Which of the following statements illustrates this?

- A. Changes are accepted up until the point that the first Sprint begins. Then, changes are deferred to a future release.
- B. Changes are accepted only if other features are removed from the backlog such that a fixed end-date is maintained.
- C. Changes are accepted up until about halfway through the project, then all changes are deferred to a future release.
- D. Changes are accepted at any time during the development effort depending on the business value of the change, the PO's acceptance, and the ability of the Scrum Team to respond in a timeframe acceptable to the PO.

Agile methodologies

-So many to choose from-



SCRUM

Set of team guidance practices, roles, events, artifacts, and rules

Based on three pillars of Transparency, Inspection, and Adaptation:

Transparency

- Visibility to those responsible for the outcome

Inspection

- Timely checks on how well a project is progressing toward goals
- Looks for problematic deviations or differences from goals

Adaptation

- Adjusting a process to minimize further issues if an inspection shows a problem or undesirable trend

Scrum Values



Focus



Respect



Courage



Openness



Commitment

Sprint



A sprint is a short, time-boxed period when a scrum team works to complete a set amount of work.



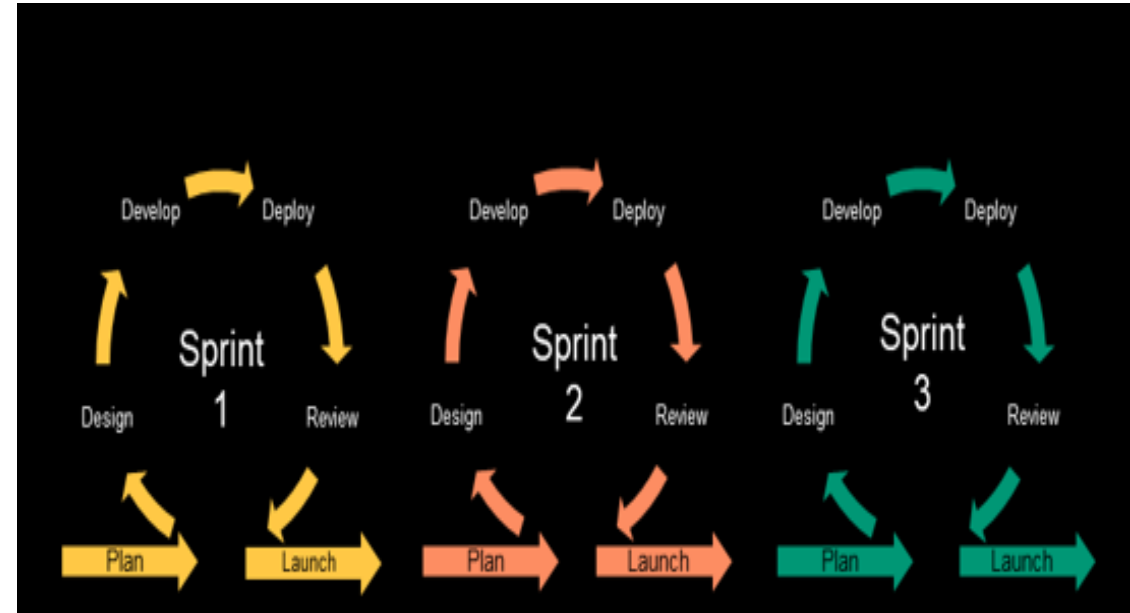
Sprints are at the very heart of scrum and agile methodologies



Sprints make projects more manageable, allow teams to ship high-quality work faster and more frequently, and gives them more flexibility to adapt to change.

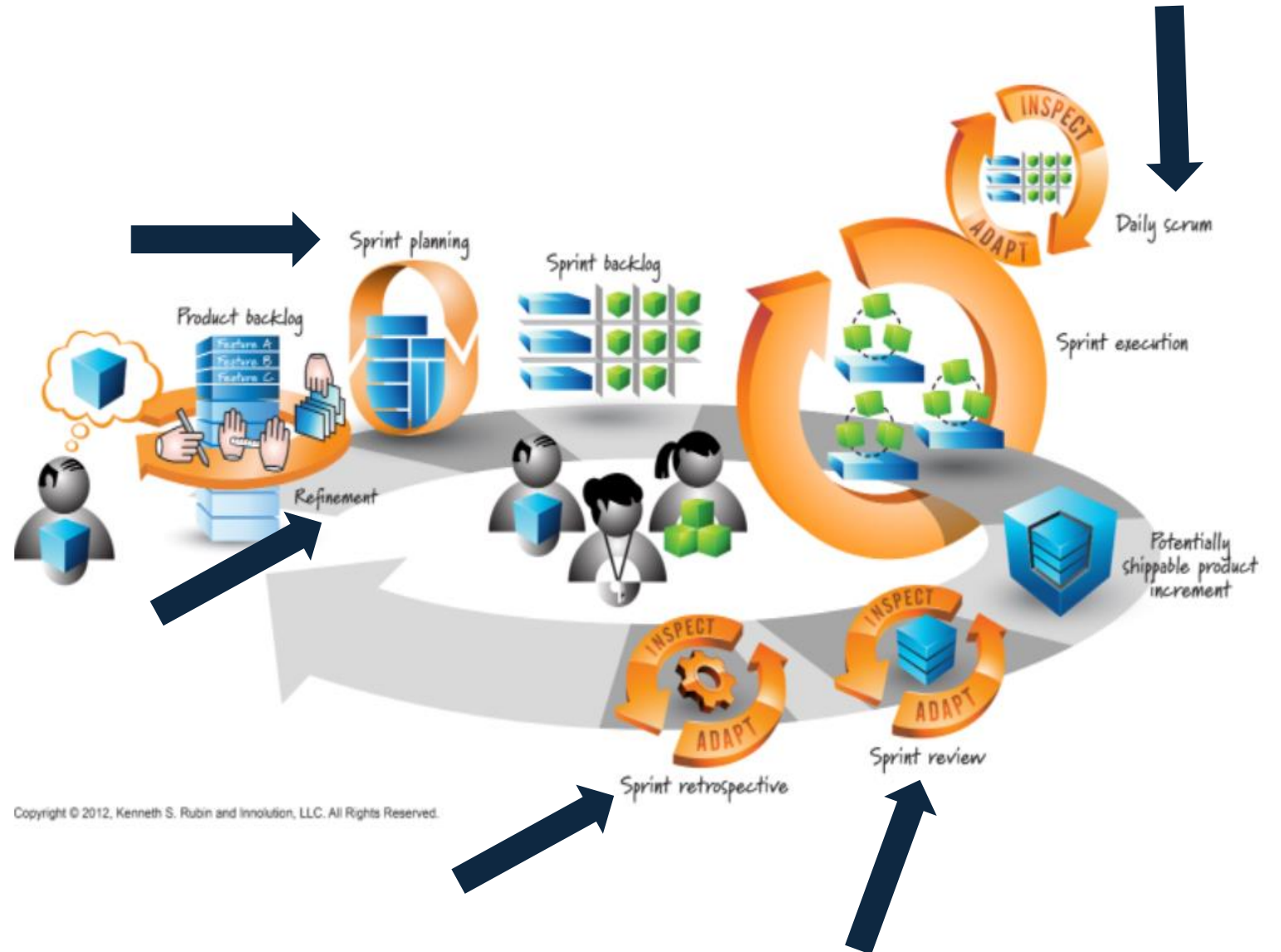
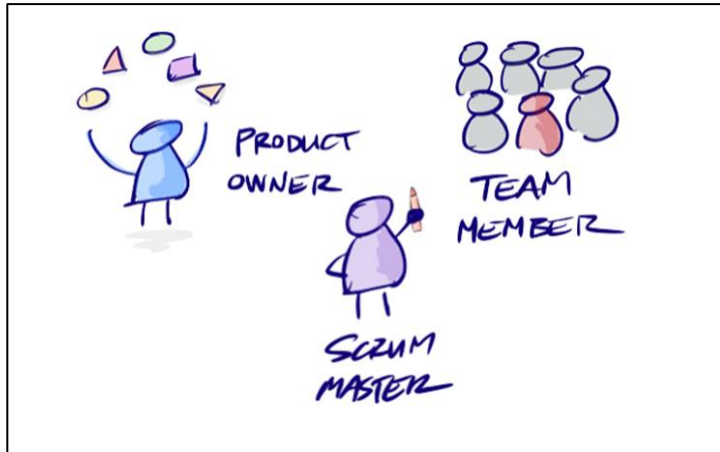


A new Sprint starts immediately after the conclusion of the previous Sprint.



Scrum Events

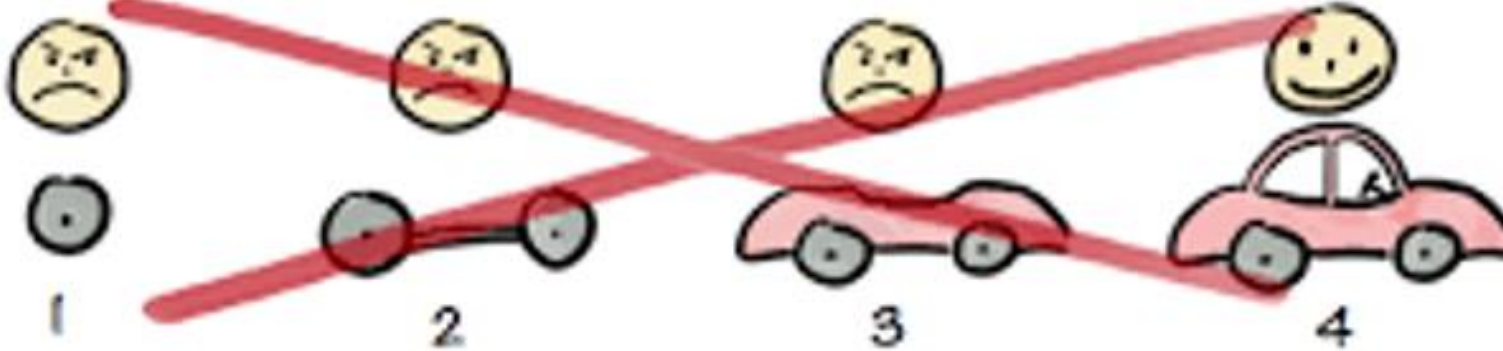
Scrum Team



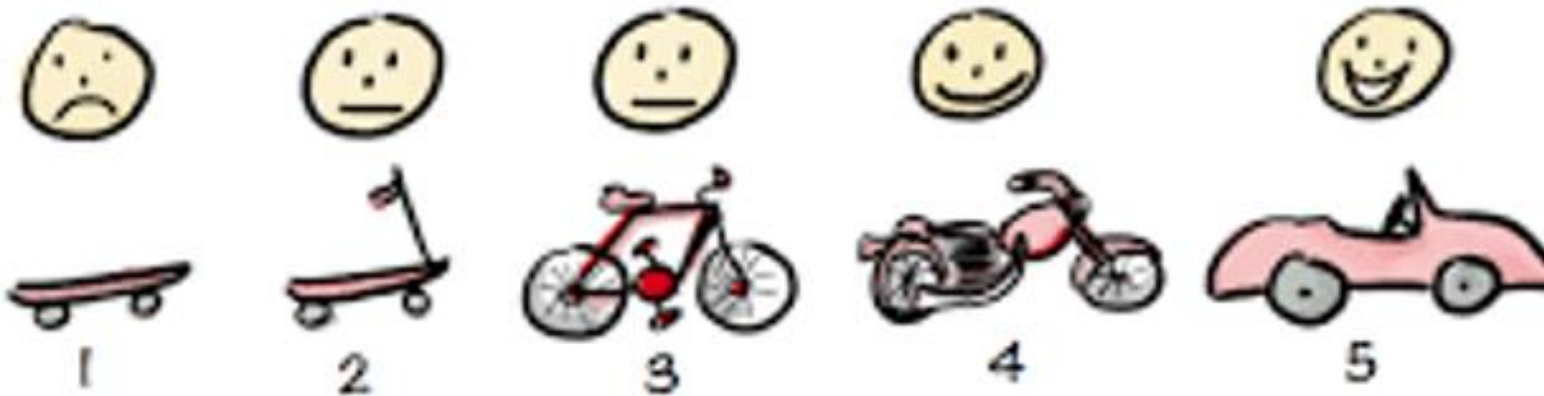
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Iterative and Incremental Delivery

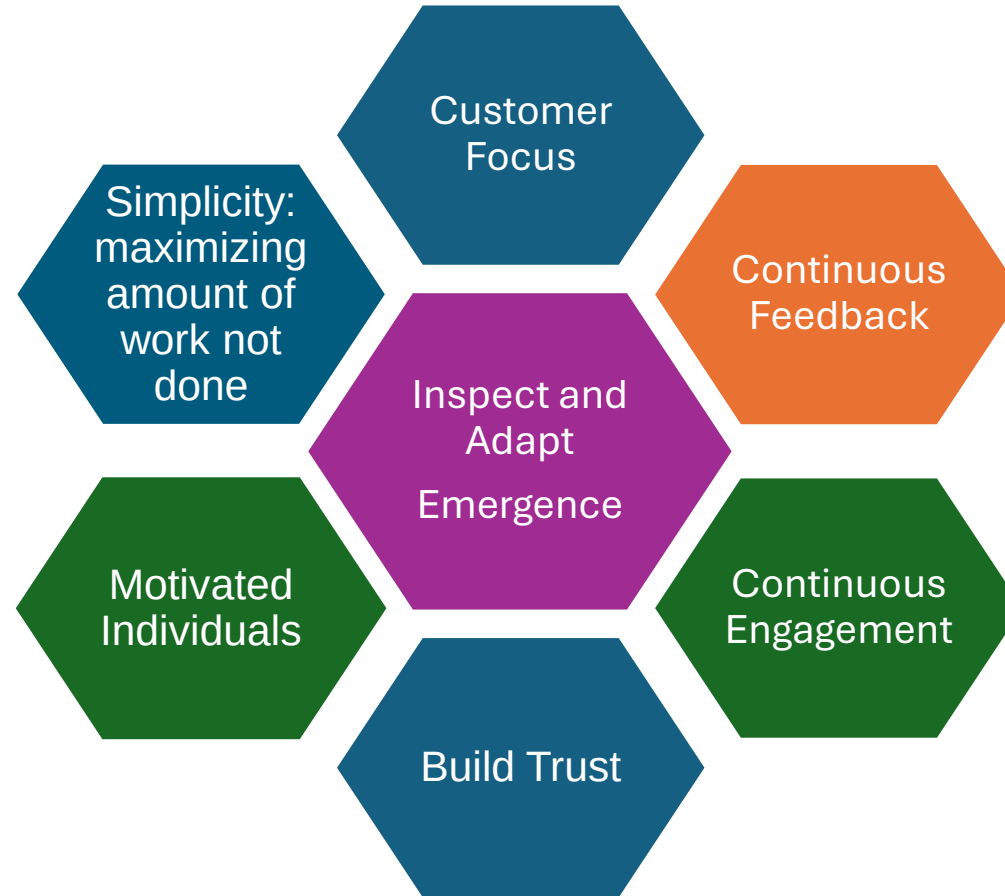
Not like this....



Like this!



Why deliver within Sprint?



Emergence implies that all solutions to all problems will become clear as we work, not simply by talking about them

Scrum Role and responsibilities

Product Owner

- Owns Product vision
- Defines features, decides on release date and content
- Responsible for market success
- Prioritizes features according to market value
- Can change features and priorities every Sprint

Scrum Master

- Responsible for facilitating process
- Focuses Team and protects them from external interruption
- Looks for ways to enhance productivity
- Assists Product Owner in leveraging Scrum

Development Team

- Small group containing all necessary project skills
- Focuses on steady delivery of high-quality features
- Generates options for delivery
- Manages own work within Sprint

Scrum artifacts

Product increment

- Part of the product that is complete after each sprint

Product Backlog

- Prioritized list of valuable items to deliver

Sprint Backlog

- List of committed items to be addressed within Sprint

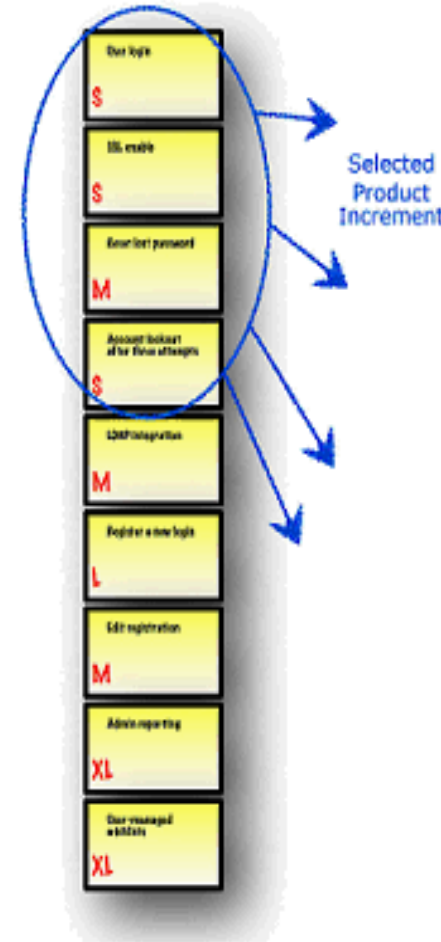
Product Backlog

Product backlog Product backlog is an ordered list of everything that might be needed in the product and is the single source of requirements for any changes to be made to the product.

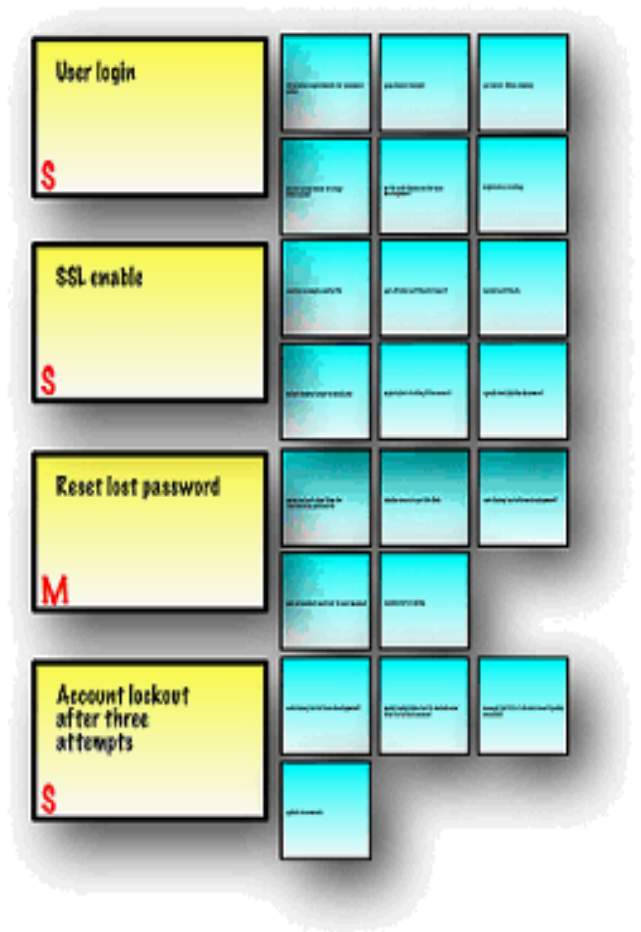
Product Backlog is used for –

- ✓ capture requests for modifying a product. This can include add new features, replacing old features, removing features and fixing issues;
- ✓ ensure the delivery team is given work which maximizes the business benefit to the owner of the product

Product Backlog

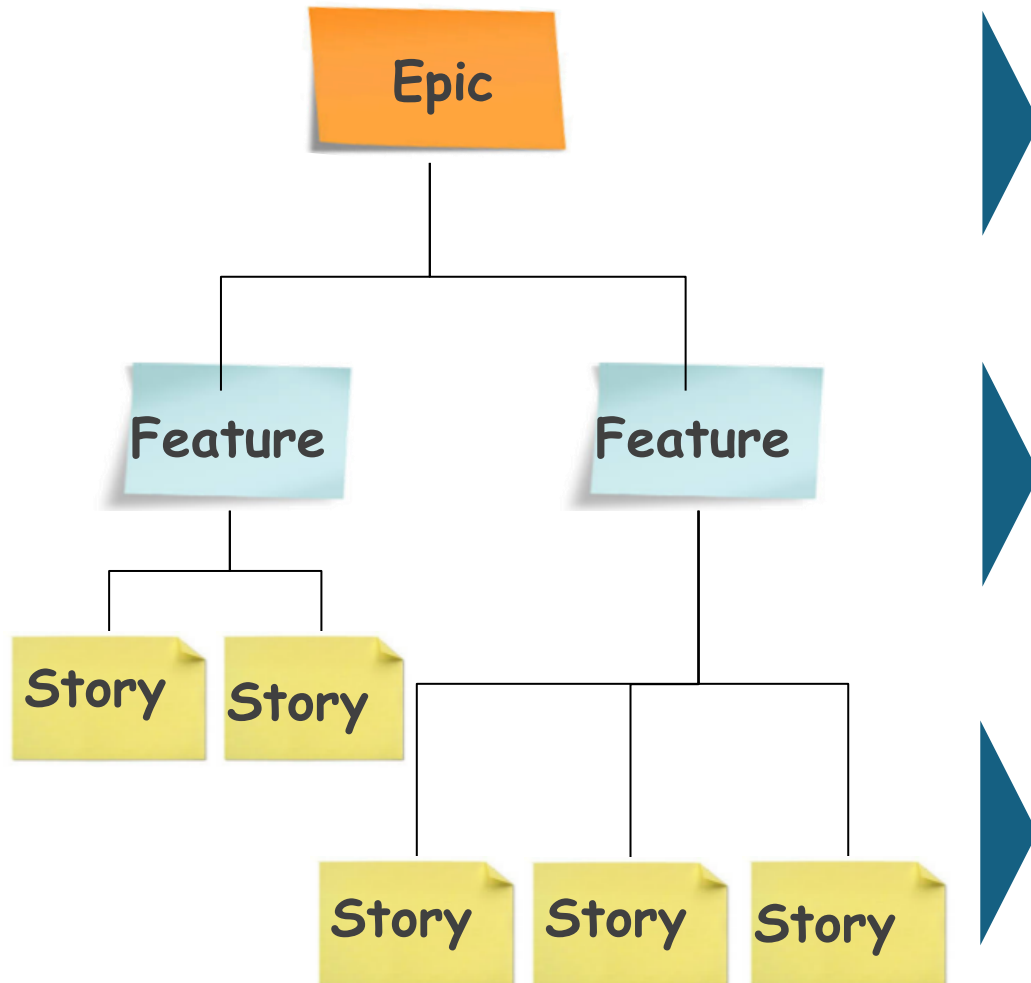


Sprint Backlog



Levels of Backlog

Requirement in the backlog are different sizes and with different levels of detail.
Epics and Features are just big User Stories waiting to be Sliced / Split.



Reimburse employees for legitimate work activities

Enter and submit expenses
Employees can enter, save and submit expenses according to our payment policies

AS AN employee
I WANT to enter my expenses and save them
SO THAT I can record expenses incurred related to work activities

User Story

- User stories are short, simple descriptions of a feature told from the perspective of the person who desires the new capability, usually a user or customer of the system.
- Describes the “what’ and not the “how”
- Description of a functionality delivering business value that can be completed within a single sprint.
- Must Include acceptance criteria

AS AN employee

I WANT to enter my expenses and save them

SO THAT I can record expenses incurred related to work activities

Acceptance criteria is the clearest and quickest way to determine whether a user story is done or not done

- Notes about what the story must do in order to accept it as complete.
- Each user story should have at least one acceptance criteria but try not to list too many.

User Story

Categories	Epics	Features	User stories
Definition	Large items which are vaguely derived from the requirements	Features are completed functionality in the product	Smallest representation of the scrum events
Example	Website for admin and user's functionality	Dashboard for the admin and users	The functionality of the dashboard to add, modify and delete users by the admin
Elementary forms	Can be converted into features and user stories	Collection of user stories forms the features. They are smaller than epic and larger than user stories	The smallest element in the product backlog which are then converted to task in the sprint backlog

Definition of Done (DOD)

Definition of Done (DoD) is a shared understanding of what it means when work is considered done, it should be defined at the beginning of the project, and it applies globally to the project.

Definition of Done (DoD) is a crucial element of a successful scrum software development

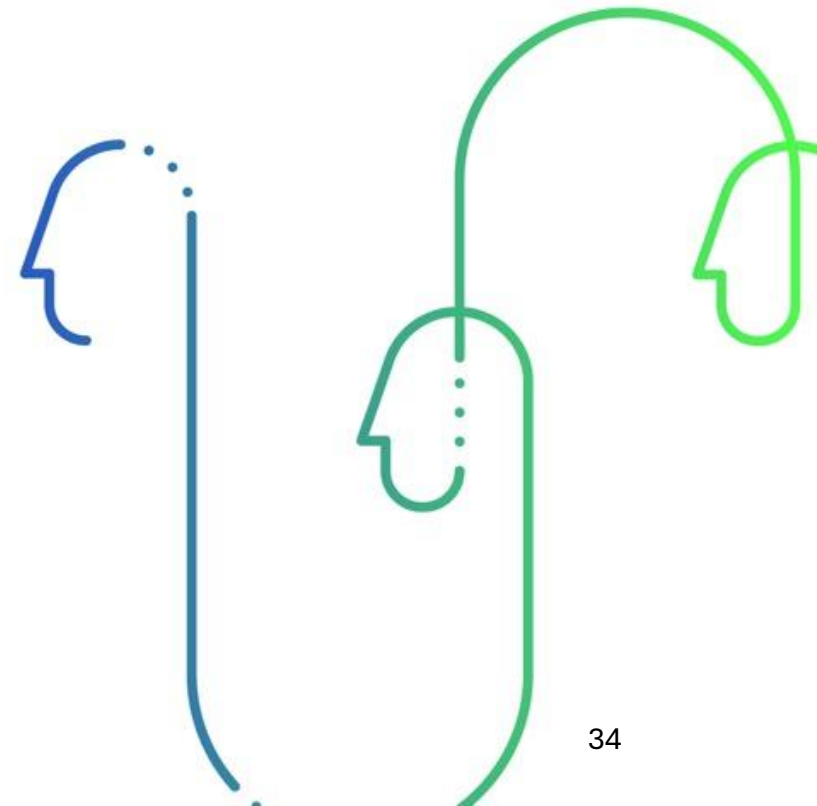
Might include things such as:

- DoD for Unit & functional tests.
- DoD Documentation.
- DoD for a Writing code

Definition of READY (DOR)

- A Definition of Ready (DoR) allows you to evaluate work before your team starts on it.
- It defines a task, user story, or story point for your team.
- This is the criteria to determine if a task or user story is ready for your team to tackle.

QUIZ



If Waterfall is plan driven, then Scrum is

- A. Client driven
- B. PO driven
- C. Business value driven
- D. none of the above

What is the ideal sprint length?

- A. 1 – 2 months
- B. 1 – 4 weeks
- C. 3 months
- D. 6 weeks

What are the scrum values?

- A. Courage, Focus, Commitment, Respect, Openness
- B. Fear, Focus, Indifference, Respect, Non-transparent
- C. Courage, Disruptive, Non-committed, Respect, Openness
- D. Courage, Focus, Commitment, Respect, Non-transparent

Which of these isn't a sprint event?

- A. Daily Stand Up
- B. Sprint Review
- C. Sprint Closure
- D. Sprint Planning

Which of the following activity is not timeboxed?

- A. Sprint
- B. Sprint Review
- C. Sprint Retrospective
- D. Product Backlog Refinement

Who should necessarily attend the Daily Standup meeting?

- A. The Scrum Team
- B. The Development Team
- C. The Scrum Team and the Stakeholders
- D. The Development Team and the Product Owner

How is Product Backlog Arranged?

- A. Items are randomly arranged
- B. Large items at the top, small items at the bottom
- C. Newer stories prioritized on top, followed by older stories
- D. High priority items on top, followed by low priority items

What are the roles in Scrum?

- A. Product Owner, Scrum Master and dev team
- B. Project Manager, Scrum Master and dev team
- C. Product Owner, manager and dev team
- D. Product Owner, Scrum Master and management

In Scrum, when is a Sprint Over?

- A. When all the Sprint Backlog tasks are completed
- B. When the final testing is completed
- C. When the time box expires
- D. When all the Sprint Backlog Items are completed

What happens when all the Sprint Items cannot be completed?

- A. The Sprint should be extended
- B. The unfinished Sprint items should be removed from the Sprint Backlog
- C. The Sprint ends with the done items
- D. Start the next Sprint with the unfinished items first